

Varun Gudibanda

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EDUCATION

University of Wisconsin-Madison

September 2020 - May 2026 (Expected)

Ph.D., Mathematics (Computational and Applied Math)

Madison, WI

Doctoral Minor in Physics

Relevant Coursework: Machine Learning in Physics, Computational Mathematics, Applied Mathematics, Fluid Dynamics, Stochastic Processes, Quantum Mechanics, Astrodynamics, Atomic Physics

University of Wisconsin-Madison

September 2020 - May 2024

M.A., Mathematics

Madison, WI

Carnegie Mellon University

August 2019 - May 2020

M.S., Mathematics

Pittsburgh, PA

Relevant Coursework: Machine Learning, Numerical Analysis, Partial Differential Equations, Finite Element Methods, Data Structures and Algorithms

Carnegie Mellon University

August 2016 - May 2020

B.S., Mathematics

Pittsburgh, PA

TECHNICAL SKILLS

Languages: Python, PyTorch, MATLAB, $\text{\LaTeX}$, Wolfram Mathematica, and Git.

RESEARCH EXPERIENCE

Graduate Student

Fall 2020-Present

University of Wisconsin-Madison

Madison, WI

- Investigating convection of fluids with internal heating.
- Analyzing the system through numerical simulations with the Dedalus library in Python.

HEDS Research Internship

May 2025-August 2025

Lawrence Livermore National Laboratory

Livermore, California

- Conducted radiation hydrodynamics simulations at the High Energy Density Science Center to design an experiment related to star formation to be performed at the National Ignition Facility

NOMAD Research Internship

June 2023-August 2023

Sandia National Laboratories

Albuquerque, New Mexico

- Implemented postprocessing effects into a workflow to reduce distortion in metal additive manufactured components

NASA Pennsylvania Space Grant Consortium

Summer 2019

Carnegie Mellon University

Pittsburgh, PA

- Developed novel energy-stable numerical methods for the Q-tensor model of liquid crystals and implemented the method in MATLAB, resulting in a publication [[arXiv:2012.00278](https://arxiv.org/abs/2012.00278)].

Summer Undergraduate Research Fellow

Summer 2018

Carnegie Mellon University

Pittsburgh, PA

- Conducted modal simulations of the inverted flag system - a cantilevered beam interacting with a surrounding fluid implemented in MATLAB, resulting in a publication [[arXiv:1910.06918](https://arxiv.org/abs/1910.06918)].
- Acquired the fellowship for \$3500 summer research funding.

AWARDS

PLS Best In Show

Summer 2025

Lawrence Livermore National Laboratory

Madison, WI

- Awarded for poster about research conducted during the HEDS internship at the Physical and Life Sciences poster session

Mathematics Exceptional Teaching Award

Spring 2024

University of Wisconsin, Madison

Madison, WI

- Awarded for demonstrating excellence in teaching in the mathematics department for the 2023-2024 academic year.

Outstanding TA Award

University of Wisconsin, Madison

2020-2021

Madison, WI

- Attained for demonstrating excellence in teaching over multiple semesters.

Outstanding Poster Award

Joint Mathematics Meetings

Spring 2019

Baltimore, MD

- Awarded for poster about research on Inverted Flag System during the Undergraduate Poster Session.

PROJECTS

Reduced Order Modeling with IRMAE-WD Autoencoder and SINDy Algorithm

Spring 2025

Personal Project

Madison, WI

- Used the Implicit Rank Minimizing Autoencoder with Weight Decay (IRMAE-WD) along with the SINDy algorithm to develop a reduced order modeling workflow
- Implemented the workflow using PyTorch and applied it to the Rayleigh-Benard Convection problem to develop a reduced order model for the system in a regime of periodic dynamics

COMMUNITY AND MENTORING

Graduate Applied Math Seminar

Fall 2023 - Spring 2025

Co-Organizer and Co-Founder

Madison, WI

- Founded a weekly seminar for graduate students in applied math as an opportunity for gaining talk and presentation experience
- Corresponded with students in the department to find speakers for the seminar and communicate with members of the seminar about seminar events

Committee on TA Policies and Procedures

Fall 2024 - Present

Member

Madison, WI

- Evaluated other teaching assistants and provide feedback and suggest points of improvement

Madison Experimental Mathematics

Spring 2024

Mentor

Madison, WI

- Mentored four undergraduate students on a fluid dynamics research project and helped them produce a talk and present at a poster session

Graduate Peer Mentoring Program

Fall 2023

Volunteer

Madison, WI

- Advised new first year graduate students entering the department about how to navigate the PhD program in terms of coursework, finding an advisor, and getting accustomed to the Madison environment

Undergraduate Mentoring Program

Fall 2022

Volunteer

Madison, WI

- Advised undergraduate students on finding research and internship opportunities as well giving graduate application advice

University of Wisconsin, Madison Math Circle

Spring 2021 - Fall 2022

Volunteer

Madison, WI

- Oversaw meetings and ensured the smooth running of talks given to engage local middle school students in mathematics.

Mega Math Competition

Spring 2022, Spring 2023

Volunteer

Madison, WI

- Formulated problems for a middle school math competition for 19 middle schools in the general Wisconsin area.

Carnegie Mellon University Math Club

Spring 2018 - Spring 2020

Secretary

Pittsburgh, PA

- Communicated with members about meetings, activities, and other mathematical opportunities.
- Helped create activities for meetings and schedule meetings.

TALKS AND POSTERS

- Radiation-Hydrodynamics Simulations to Increase Turbulent Mach Number in Discovery Science**
Star Formation Experiments at the National Ignition Facility
Physical and Life Sciences Poster Session Summer 2025
Livermore, CA
- A Distortion Compensation Workflow Accounting for the Effects of Post-Processing in Metal Additive Manufacturing**
TMS Annual Meeting and Exhibition Spring 2024
Orlando, FL
- Rayleigh-Bernard Convection**
Carnegie Mellon University Math Club Spring 2022
Pittsburgh, PA
- Dynamics of the Inextensible Inverted Flag with Piston-Theoretic Forcing Term**
Joint Mathematics Meetings Spring 2019
Baltimore, MD
- 4 Talks at Graduate Applied Mathematics Seminar** Fall 2023, Spring 2024, Fall 2024, Spring 2025

CONFERENCES AND WORKSHOPS

- Joint Mathematics Meetings 2025** January 2025
Baltimore, MD
- TMS Annual Meeting and Exhibition 2024** March 2024
Orlando, FL
- Fluid Mechanics of Planets and Stars** April 2023
Udine, Italy
- Joint Mathematics Meetings 2019** January 2019
Baltimore, MD

PUBLICATIONS

- Hexagonal coherent structures in Rayleigh Bénard and radiatively driven convection** (with Leslie Smith and Fabian Waleffe) *In Preparation*
- Convergence analysis of a fully discrete energy-stable numerical scheme for the Q-tensor flow of liquid crystals** (with Franziska Weber and Yukun Yue), *SIAM Journal on Numerical Analysis* (2022)
- Large deflections of inextensible cantilevers: modeling, theory, and simulation** (with Maria Deliyianni, Jason Howell, Justin T. Webster), *Mathematical Modelling of Natural Phenomena* (2020)

MISCELLANEOUS

- Former Top 200 in US at Rubik's Cube Blindfolded** Summer 2016
US Nationals Rubik's Cube Competition Portland, OR